

Chapter 2 - Literature Review

CHAPTER 2

LITERATURE REVIEW

2.1 Literature Review 1 — Cooling Load Calculation Principles

Cooling load is the amount of heat that must be removed from room air to maintain the required indoor temperature. Determination of cooling load requires estimation of heat transmission into the space through the building envelope and consideration of the three modes of heat transfer: conduction, convection and radiation. The calculation also requires consideration of thermal properties, outdoor design conditions, infiltration and internal heat gains.

For non-residential buildings, cooling-load calculation methods are based on determining the heat gains associated with the building and converting these heat gains into the cooling load of the conditioned space. Important sources include heat transfer through building surfaces, solar effects, infiltration and internal heat gains from occupants, lighting and equipment. The time-dependent behaviour of heat transfer is important because heat entering a building component may be stored within its thermal mass before being released to the space.

The heat-balance approach considers the interaction between the different heat-transfer processes at the surfaces and within the conditioned space. Mathematical modelling is used to determine the resulting cooling requirement. These principles provide the basic foundation for determining the cooling capacity required for an air-conditioned space.

2.2 Literature Review 2 — Development of Building Peak Cooling Load Methods

Accurate sizing of HVAC systems requires an appropriate method for determining the peak cooling load. Several cooling-load calculation methods have been developed for building applications. The major methods published by ASHRAE include the Total Equivalent Temperature Difference/Time Averaging Method (TETD/TA), Transfer Function Method (TFM), Cooling Load Temperature Difference/Solar Cooling Load/Cooling Load Factor Method (CLTD/SCL/CLF), Heat Balance Method (HBM), and Radiant Time Series Method (RTSM).

The development of these methods represents a progression from simplified approaches toward methods that more directly represent the thermal behaviour of buildings. The Heat Balance Method considers heat balances at building surfaces and within the conditioned space, while the Radiant Time Series Method provides a simplified procedure derived from the heat-balance approach.

Differences in the assumptions and treatment of heat transfer in these methods can produce differences in the calculated sensible cooling load. Therefore, the selection and proper application of the calculation method are important when determining design peak cooling loads and selecting HVAC equipment.

2.3 Literature Review 3 — Heat Balance / Radiant Time Series Methods

The Radiant Time Series (RTS) method is a design cooling-load calculation procedure derived from the Heat Balance Method. It provides a simplified method for accounting for the time-dependent relationship between heat gain and cooling load. The method uses a 24-term response-factor series to calculate conductive heat gain and a 24-term radiant time series to convert instantaneous radiant heat gain into cooling load.

Heat gains entering a conditioned space are separated into radiant and convective components. The convective portion contributes directly to the cooling load, while the radiant portion is absorbed by surfaces and subsequently released to the space over time. Consequently, the radiant heat gain occurring at a particular hour does not necessarily become cooling load entirely during that same hour.

The radiant time factors are used to determine the fraction of radiant heat gain that becomes cooling load during the current and subsequent hours. The procedure therefore accounts for the thermal response of the building and the delay between heat gain and its effect on room cooling load.

2.4 Literature Review 4 — Cooling Load Calculation in Different Indoor Conditions

Occupancy is an important component of the internal cooling load of a building. Traditional mechanical design may use an assumed occupancy number that can be higher than the actual occupancy. When actual occupancy is lower than the design assumption, the calculated internal heat gain and resulting cooling load may be overestimated.

A case study of an office building in Tianjin considered occupancy patterns together with the heat gains from occupants, lighting and equipment. In the studied building, these interior heat gains represented 66.6% of the total cooling load, while the chiller operated at a part-load ratio of only 30%. An occupancy-based approach using a Markov transition matrix was developed from occupancy survey data.

A correction coefficient was proposed for design-stage cooling-load calculation to account for occupancy conditions. For the case studied, this approach resulted in a 35.9% reduction in calculated cooling load compared with the conventional design calculation. The study demonstrates that actual indoor conditions, particularly occupancy behaviour, can have a significant influence on calculated cooling requirements.

2.5 Literature Review 5 — Effect of Uncertainty on Cooling Load Calculation

Peak cooling load calculations are generally performed during the early stages of building design, when complete information about the building and its operating conditions may not be available. Uncertainty can arise from lack of information, the approximate nature of the building model and random or variable factors such as occupancy and weather conditions.

The calculated cooling load can therefore depend considerably on the assumptions used for input parameters. Conventional HVAC design commonly deals with these uncertainties through worst-case conditions and safety factors. However, applying safety factors at different stages can lead to significantly different load estimates and may result in oversized equipment.

A stochastic approach can be used to represent uncertainty by assigning uncertainty bands to input parameters and propagating them through the building model. Instead of producing only one cooling-load value, this approach can produce a probability distribution representing a range of possible peak cooling loads. Sensitivity analysis can then be used to identify the input parameters that have the greatest influence on the calculated result.

This highlights the importance of using accurate and representative building data in practical cooling-load calculations.

2.6 Literature Review 6 — Practical Application of Cooling Load Calculation for HVAC Design

Design cooling-load calculation is an important part of HVAC system design because the calculated load is used to determine the required capacity of air-conditioning equipment. Different calculation methods may produce different results because they use different assumptions and approaches for representing building heat transfer.

A quantitative comparison was carried out between the ASHRAE Heat Balance Method, Radiant Time Series Method and the U.K. Admittance Method. The comparison considered parameters such as exterior wall type, roof type, glazing area and weather conditions. The calculated peak loads from simplified methods were compared with those obtained using the Heat Balance Method.

The results showed general trends of overprediction or underprediction of peak loads by simplified methods when compared with the Heat Balance Method. The differences were related to the assumptions used within the calculation procedures and the way important heat-transfer mechanisms were represented.

The comparison demonstrates the practical importance of selecting and applying an appropriate cooling-load calculation procedure when determining HVAC equipment requirements.

2.7 Summary of Literature Review

The reviewed literature establishes the theoretical and practical basis for cooling-load calculation. The fundamental principles show that cooling load is determined by considering heat transfer through the building envelope, solar effects, infiltration and internal heat gains. Different peak cooling-load methods have been developed, including TETD/TA, TFM, CLTD/SCL/CLF, HBM and RTSM, with differences in their assumptions and treatment of heat transfer.

The Heat Balance and Radiant Time Series approaches demonstrate the importance of considering the time-dependent behaviour of heat gain. Occupancy-related research further shows that actual indoor conditions can substantially affect the calculated cooling requirement. Uncertainty studies indicate that incomplete or inaccurate input data can influence the final peak-load result, while comparative studies demonstrate that the selected calculation procedure can affect HVAC design results.

Overall, the literature indicates that reliable cooling-load estimation requires consideration of building dimensions and construction, outdoor conditions, solar radiation, occupancy, lighting, equipment, ventilation/infiltration and the appropriate calculation method. These findings provide the technical foundation for the present project, in which actual measurements and operating conditions of the HOD cabin will be used to determine its cooling load.